



University of Central Arkansas

Department of Computer Science and Engineering

Fall 2026 – CSEC 2300 Introduction to Cybersecurity

Instructor:

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Office Phone: 501-852-2050

Office: MCS 313

Course:

Title: INTRODUCTION TO CYBERSECURITY

Course Number: 2300

CRN (Course Reference Number): 23330

Website: Brightspace

Lecture Times MWF – 08:00 AM-08:50 AM

Lecture Location: Room 339

Office Hours: Monday and Wednesday, 3:00–5:00 PM, or by appointment via email

Office hours are subject to change based on university committee responsibilities

Course Description:

This course provides an introduction to cybersecurity. It covers fundamental concepts necessary to understand the threats to security as well as various defenses against those threats. The material includes an understanding of existing threats, planning for security, technology used to defend a computer system, and implementing security measures and technology. A required course for Cybersecurity majors.

Prerequisite: CSCI 1470 COMPUTER SCIENCE I

Course Objectives:

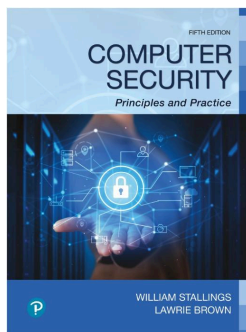
Upon successful completion of this course, you will be able to:

1. **Explain** fundamental cybersecurity concepts, including the CIA triad, threats, vulnerabilities, and risk.
2. **Identify and analyze** common cyber threats and attack techniques, such as malware, phishing, and denial-of-service attacks.
3. **Apply** basic security mechanisms, including encryption, authentication, and access control, to protect systems and data.
4. **Assess and manage risk** by identifying vulnerabilities and proposing appropriate mitigation strategies.

Textbook:

Required: There is no required textbook for this course. The instructor will provide the required reading materials.

Recommended: (Not mandatory) Computer Security Principles and Practice, 5th edition, William Stallings and Lawrie Brown



Tentative Topics and Schedule:

This schedule is tentative and may be adjusted during the semester based on class progress, instructional needs, or other circumstances. Any changes will be communicated to students in advance.

Week/Date	Topics	Assignment
Week 0 (Aug 21)	Course Introduction Syllabus overview	
Week 1 (Aug 24, 26, 28)	Module 1: Introduction to Cybersecurity	Quiz 1 Assignment-1 Released

	Topics Cybersecurity overview, CIA triad, threats, Vulnerabilities and Attacks	
Week 2 (Aug 31, Sep 2, 4)	Module 2: Core Security Mechanisms	Quiz 2
	Topics User Authentication Password Security Multi-Factor Authentication Access Control Authorization and Accountability Logging, Monitoring, & Auditing	
Week 3 (Sep 7, 9, 11) (Labor Day – no class on Sep 7)	Module 3: Cryptography (Part 1)	Assignment-1 Due (Sept 11)
	Topics Cryptographic Tools Symmetric Encryption Message Confidentiality	
Week 4 (Sep 14, 16, 18)	Module 3: Cryptography (Part 2)	Quiz 3 Assignment-2 released: Cryptography Lab (CrypTool)
	Topics Public-Key Cryptography Message Authentication Key Exchange	
Week 5 (Sep 21, 23, 25)	Module 3: Cryptography (Part 3)	Quiz 4
	Topics Hash Functions Digital Signatures	
Week 6 (Sep 28, 30, Oct 2)	Module 4: System Security	Quiz 5
	Topics Operating System Security Operating systems hardening Patch Management	Assignment-2 Due (Oct 2)
Week 7 (Oct 5, 7, 9)	Module 5: Attacks and Threats (Part 1)	Quiz 6
	Topics: Malware Social Engineering Spam E-mail, Trojan horses Ransomware Phishing and identity theft	
Week 8 (Oct 12, 14)	Revision, Reading and Midterm Exam	Midterm Exam: Oct 14

	Oct 15–18 — Fall Break (No Class)	
Week 9 (Oct 19, 21, 23)	Module 5: Attacks & Threats (Part 2)	Assignment-3 released(SQL Injection)
	Topics: Web Application Security SQL Injection	
Week 10 (Oct 26, 28, 30)	Module 5: Attacks & Threats (Part 3)	Quiz 7
	Topics: Cross-Site Scripting (XSS) Denial-of-Service Attacks	Assignment-4 Released (XSS)
<u>Week 11 (Nov 2, 4, 6)</u>	Module 6: Network Security & Defense I	Assignment-3 Due (Nov 4)
	Topics: Network Security Fundamentals TCP SYN flooding attack	Quiz 8
<u>Week 12 (Nov 9, 11, 13)</u>	Module 6: Network Security & Defense II	Assignment-4 Due (Nov 13)
	Topics Intrusion Detection Systems (IDS) Intrusion Prevention Systems (IPS) Nmap and Wireshark Demonstration, open ports	Quiz 9 Assignment-5 Released
<u>Week 13(Nov 16, 18, 20)</u> <u>Week 14 (Nov 23)</u>	Module 7: Security Management, Risk, and Governance	Quiz 10
	Topics IT Security Management and Risk Assessment IT Security Controls, Plans, and Procedures Legal and Ethical Aspects of Cybersecurity	
<u>Week 15 (Nov 30, Dec 2)</u>	Course Review & Final Exam Preparation (Dec 4: Study Day (No Class))	Assignment-5 Due (Dec 2)
<u>Dec 5–10</u>	Final Exam (exact date assigned by the university) Dec 9, Time 8:00 AM - 10:00 AM	
<u>December 14</u>	Grades Due	

Summary

- **10 Quizzes**
- **5 assignments/labs**
- **Midterm**
- **Final exam**

Evaluation (%):

Assignments/Labs	30 %
Quizzes	25 %
Midterm	20 %
Final	25 %
TOTAL	100 %

A note about evaluation: Each question in the quizzes carries equal weight. Your overall quiz grade is calculated by dividing the total points earned across all quizzes by the total points available. Assignments follow a similar structure. While some assignments may be worth 100 points and others 150 points, your assignment grade is calculated as the total points you earned divided by the total points available from all assignments.

Grading Scale:

A	90 - 100
B	80 - 89.99
C	70 - 79.99
D	60 - 69.99
F	0 - 59.99

Assignments: There will be 5 individual assignments/labs throughout the semester. To receive full credit, you must complete the assignments by the due dates. Late submission penalty is 10% of the assignment weight per calendar day. However, you will receive no credit for an assignment that is more than three calendar days late. To ensure fairness, no extra credit assignments will be given.

Valid reasons for missing a deadline: (1) School related functions (band, orchestra, sports events, etc.) (2) Medical reasons (illness, surgery, etc.) or accidents (e.g., motor vehicle) (3) Family emergency reasons (illness, surgery, death, etc.) in immediate family (parents, spouses, siblings, and children). In any case, prior contact with the instructor is required. Also, proper documentation (note from the coach, instructor, supervisor, physician, police, etc.) within three days is required. If your excuse is deemed to be valid, you will not be penalized for late submission for a maximum of three days.

Quizzes: There will be several (10) quizzes during the semester. Some quizzes will be administered in class to assess comprehension and may also be used to document attendance. Students may attempt each online quiz up to three times. The highest score among all attempts will be recorded as the final score for that quiz

Important Dates

Please check the Academic Calendar at: <https://uca.edu/registrar/academic-calendar>.

Event	Date
Classes Begin	Aug 20 (First Class Aug 21)
Labor Day	Sep 7
Fall Break	Oct 15–18
Thanksgiving Break	Nov 25–29
Study Day	Dec 4
Final Exams	Dec 5–10
Grades Due	Dec 14

<https://uca.edu/registrar/files/2026/01/FINAL-EXAMINATIONS-2026-FALL-SEMESTER.pdf?1>

Office Hours:

Office hours are face-to-face. Zoom can be used for convenience. If you need to see me at another time, please email me to schedule a meeting.

Resources:

- Syllabus/Notes (Slides): Electronic copies will be made available on Brightspace.
- UCA Email: All email communication will be done via UCA email and Brightspace. You are required to check your email every day as information such as the availability of specifications, submission deadlines, and changes to the course schedule will be communicated via email.

Course Plan:

- The lectures will be in class.

- In case the classes need to be held online, ZOOM will be the environment, and the links will be posted in advance.
- I can be reached at the UCA e-mail address radhikari2@uca.edu.
- If you must miss class for a valid reason, please notify the instructor as early as possible. Students are responsible for making up missed course material and activities.

Classroom Conduct

The following rules are to ensure a classroom environment that is conducive to learning:

- Please silence or mute any potentially disruptive electronic devices (pagers, mobile phones, portable media players, mobile computing devices, etc.)
- Please do not engage in private conversations during lectures.
- Please do not engage in non-class-related activities such as listening to music, browsing, playing games, etc.

Weekly Effort for the Class

On average, 6-10 hours of study per week will bring you to a good level of understanding of the concepts discussed in the classroom.

Academic Integrity Policy

The University of Central Arkansas affirms its commitment to academic integrity and expects all members of the university community to accept shared responsibility for maintaining academic integrity. Students in this course are subject to the provisions of the university's Academic Integrity Policy, approved by the Board of Trustees as Board Policy No. 709 on February 10, 2010, and published in the *Student Handbook*. Penalties for academic misconduct in this course may include a failing grade on an assignment, a failing grade in the course, or any other course-related sanction the instructor determines to be appropriate. Continued enrollment in this course affirms a student's acceptance of this university policy.

Americans with Disabilities Act Statement

The University of Central Arkansas adheres to the requirements of the Americans with Disabilities Act. If you need an accommodation under this Act due to a disability, please contact the Office of Accessibility Resources and Services (OARS), 450-3613.

Building Emergency Plan Statement

An Emergency Procedures Summary (EPS) for the building in which this class is held will be discussed during the first week of this course. EPS and Building Emergency Plan (BEP) documents for most buildings on campus are available at <https://uca.edu/go/bep-library>. Every student should be familiar with emergency procedures for any campus building in which he/she spends time for classes or other purposes.

The Title IX Disclosure

In furtherance of its core values— academic vitality, integrity, and diversity—UCA is dedicated to promoting a campus community free from discrimination. Title IX of the Education Amendments Act of 1972 requires all educational institutions to address gender-based discrimination on campus, and UCA implements these Federal requirements through a fair, consistent, and appropriate process of investigation and adjudication. Please see UCA's Title IX website (<https://uca.edu/titleix/>) for the university's policy, relevant forms, training opportunities, and related information.

Attendance and Drop Policy:

Regular class attendance is expected and strongly encouraged. Some in-class quizzes and activities may also serve as attendance records. Students who wish to withdraw from the course are responsible for following the University's official withdrawal procedures and deadlines.

Other Policies

Students should familiarize themselves with all policies included in the Student Handbook, particularly the following:

- Sexual Harassment Policy
- Academic Policies

AI Technologies

AI tools may be used as **learning aids** to help understand concepts and review material. However, students are responsible for the accuracy of their work and **must be able to explain, reproduce, and justify anything they submit**. The instructor may ask students to demonstrate their understanding of submitted work.

AI tools are not permitted during quizzes or exams unless explicitly authorized by the instructor. Unauthorized AI use may be treated as an academic integrity violation.

Student Evaluations

Student evaluations of a course and its professor are a crucial element in helping faculty achieve excellence in the classroom and the institution in demonstrating that students are gaining knowledge. Students may evaluate courses they are taking starting on the Monday of the thirteenth week of instruction through the end of finals week by logging in to myUCA and clicking on the Feedback Hub.

A final note:

If you have a concern about this class, please contact the instructor/me first. If you would rather talk to someone else, please contact the department chair (ecelabi@uca.edu).